

Vanier College – Advanced Networks

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Subnetting: Multi-Department Network

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Before you start: build it like a professional

Set up your topology the way a network engineer would document it, before assigning any IP addresses:

- Rename devices with meaningful hostnames (R-HQ, R-Remote, SW-Sales, PC-Sales-01) instead of Packet Tracer defaults - click the name label, or use Config > Global Settings > Display Name.
- Label departments on the canvas (Place Note tool) if hostnames alone aren't clear enough.
- Use Copper Straight-Through for PC-to-switch and switch-to-router links, and Fiber for the router-to-router WAN link if using SFP fiber modules.
- Any switch model works here - no VLANs or advanced config needed.
- Recommended router: Cisco 2911 (3 onboard GigabitEthernet ports covers the 3 departments). For the 4th interface (WAN link), power the router off, add an HWIC-1GE-SFP in an empty HWIC slot, seat a GLC-LH-SMD in its SFP cage, then power back on.

Scenario

You are given 192.168.10.0/24 to divide among three department LANs on R1, one remote-site LAN on R2, and a point-to-point fiber link between R1 and R2. Departments have different host counts, so size each subnet to fit - largest first.

Given requirements

This is typically all the information you'd get from a client or manager: departments, rough headcounts, which sites connect. No addresses or masks - turning this into an addressing plan is your job in Step 1.

Segment	Hosts needed	Connects to
Sales LAN	up to 50	R1 (R-HQ)
Engineering LAN	up to 20	R1 (R-HQ)
Management LAN	up to 10	R1 (R-HQ)
Remote LAN	up to 5	R2 (R-Remote)
R1-R2 WAN link	2 (point-to-point)	R1 & R2

Step 1: Subnet the address block

Fill in every column for each segment, largest requirement first.

Segment	Network address	CIDR	Subnet mask	First usable host	Last usable host	Broadcast address	Usable hosts
Sales	192.168.10.0	/26	255.255.255.192	192.168.10.1	192.168.10.62	192.168.10.63	62

Segment	Network address	CIDR	Subnet mask	First usable host	Last usable host	Broadcast address	Usable hosts
Engineering	192.168.10.64	/27	255.255.255.24	192.168.10.65	192.168.10.94	192.168.10.95	30
Management	192.168.10.96	/28	255.255.255.40	192.168.10.97	192.168.10.110	192.168.10.111	14
Remote	192.168.10.112	/29	255.255.255.48	192.168.10.113	192.168.10.118	192.168.10.119	6
WAN link (R1-R2)	192.168.10.120	/30	255.255.255.52	192.168.10.121	192.168.10.122	192.168.10.123	2

Step 2: Assign router interface addresses

No default-gateway column here - a router doesn't need one. A gateway is for a device that can't route and hands off anything unfamiliar to one trusted device. A router keeps its own routing table instead, and adds each configured interface's subnet to it automatically as a 'connected' route.

Router interface	IP address	Subnet mask
R1 (R-HQ) - Sales interface	192.168.10.1	255.255.255.192
R1 (R-HQ) - Engineering interface	192.168.10.65	255.255.255.224
R1 (R-HQ) - Management interface	192.168.10.97	255.255.255.240
R1 (R-HQ) - WAN interface	192.168.10.121	255.255.255.252
R2 (R-Remote) - Remote interface	192.168.10.113	255.255.255.248
R2 (R-Remote) - WAN interface	192.168.10.122	255.255.255.252

Step 3: Assign PC addresses

PCs can't route, so they need a default gateway: the router interface on their own subnet. Sales PCs use R1's Sales interface (192.168.10.1); Remote PCs use R2's Remote interface (192.168.10.113); and so on.

Device	IP address	Subnet mask	Default gateway
Sales - PC1	192.168.10.2	255.255.255.192	192.168.10.1
Sales - PC2	192.168.10.3	255.255.255.192	192.168.10.1
Engineering - PC1	192.168.10.66	255.255.255.224	192.168.10.65
Engineering - PC2	192.168.10.67	255.255.255.224	192.168.10.65
Management - PC1	192.168.10.98	255.255.255.240	192.168.10.97
Management - PC2	192.168.10.99	255.255.255.240	192.168.10.97
Remote - PC1	192.168.10.114	255.255.255.248	192.168.10.113
Remote - PC2	192.168.10.115	255.255.255.248	192.168.10.113

Checkpoint: test before you fix it

Configure everything above, then ping a Remote PC from a Sales PC. It will fail. Before reading on, write down why - think about what Step 2 said about how a router learns its routes.

Step 4: Static routes needed

A router only learns about its own directly connected networks automatically - it has no way to discover anything further away, even across a link to another router. Since no dynamic routing protocol is running here, each router needs to be manually told: 'this network isn't mine, but send matching traffic to this next hop.' That's what fixes the checkpoint failure.

Router	Destination network	Mask	Next hop
R1	192.168.10.112	255.255.255.248	192.168.10.122
R2	192.168.10.0	255.255.255.192	192.168.10.121
R2	192.168.10.64	255.255.255.224	192.168.10.121
R2	192.168.10.96	255.255.255.240	192.168.10.121

Step 5: Fundamental questions

You should be able to answer these well enough to rebuild this lab from scratch, with different numbers, unaided.

1. Why is it useful to divide a single large address block into several smaller subnets instead of one flat network?

It allows better organization. It also allows you to optimize your networks to avoid waste. For example, it would be a waste to have a network that allows you to have 100 available hosts when you only 2

2. Why size the largest requirement first when subnetting for several different-sized needs?

Because it is easier determine the amount of space needed for each subnet

3. What is the difference between a network address and a broadcast address in the same subnet?

The network address identifies the subnet itself

The broadcast address allows you to communicate with devices inside that subnet

4. Why does a router interface not need a default gateway, while a PC does?

Because a router already contains a routing table to point to other devices. A PC needs one in order to communicate with other devices

5. A ping from Sales to Remote failed even though every device had correct addressing. Why?

Because it cannot detect it. Even if both routers are connected, they are far away from each other. To counteract this we need to hop, tell them the exact path they need to take.